
EE1060: Differential Equations and Transform Techniques

Lecture 31: Laplace Transform

Topics :

1. Motivation
 2. Existence and Region of Convergence of Laplace Transform
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Motivation - Computing System Response

Response of an LTI system for e^{st} :-

$$s \in \mathbb{C}$$

$$y(t) = e^{st} * h(t) = h(t) * e^{st}$$

$$s = \sigma + j\omega$$

$$= \int_{-\infty}^{\infty} h(\tau) e^{s(t-\tau)} d\tau$$

$$= e^{st} \underbrace{\int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau}_{H(s)}$$

$H(s) \rightarrow$ Does it converge?

$$= e^{st} \underline{H(s)}$$

- a) $\underline{e^{st}} \rightarrow$ eigen function of a LTI system $\Rightarrow e^{\sigma t} \underbrace{e^{j\omega t}}_{\sin/\cos}$ when $\sigma=0$
- b) $\underline{H(s)} \rightarrow$ eigen value.

if $x(t) = \sum_i K_i e^{s_i t}$, then $y(t) = \sum_i K_i e^{s_i t} H(s_i)$

$$K_i e^{s_i t} \leftrightarrow K_i e^{s_i t} \underline{H(s_i)} \leftarrow$$

Provided it exists?

Motivation - Computing System Response

Example:- $\frac{dy}{dt} + ay = x(t) \Rightarrow$ impulse response $= e^{-at} u(t)$

$$x(t) = \cos \omega_0 t$$

Step 1:-

$$\begin{aligned} H(s) &= \int_{-\infty}^{\infty} h(t) e^{-st} dt \\ &= \int_0^{\infty} e^{-(s+a)t} dt = \left. -\frac{1}{s+a} e^{-(s+a)t} \right|_0^{\infty} \\ &= \frac{-1}{s+a} e^{-(\sigma+a)t} e^{-j\omega t} \bigg|_0^{\infty} \end{aligned}$$

when $\sigma + a > 0 \rightarrow H(s)$ to be defined
 $\Rightarrow \sigma > -a$
 $\Rightarrow \text{Re}\{s\} > -a$ $= \frac{1}{s+a}$

Step 2:- $x(t) = \frac{1}{2} (e^{j\omega_0 t} + e^{-j\omega_0 t})$
 $s_1 = j\omega_0 \quad s_2 = -j\omega_0$

$H(s_1)$ & $H(s_2)$ are defined.

Step 3:

$$y(t) = \frac{1}{2} \underbrace{H(s_1)} e^{j\omega_0 t} + \frac{1}{2} \underbrace{H(s_2)} e^{-j\omega_0 t}$$

Extending the idea to Signals - Laplace Transform

Extending the definition to a signal $x(t)$ is general,

$$X(s) = \mathcal{L}\{x(t)\} = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

Region of Convergence (ROC): Values of s for which the Laplace T/F of a signal is defined / converges.

if $s = j\omega$,
$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = F\{x(t)\} \rightarrow \text{if } j\omega \in \text{ROC of LT}$$

$$X(j\omega)$$

Another condition for existence:-

$$X(s) = \int_{-\infty}^{\infty} \underbrace{x(t) e^{-\sigma t}} e^{-j\omega t} dt$$

Laplace T/F is Fourier T/F of Signal $x(t) e^{-\sigma t}$

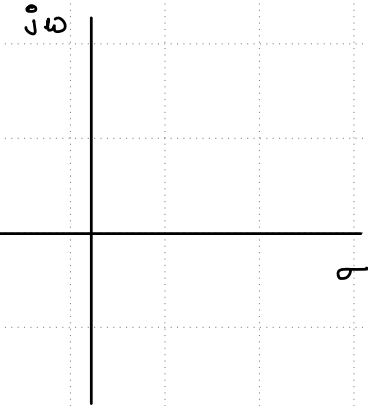
$$\text{Condition} = \int_{-\infty}^{\infty} |x(t) e^{-\sigma t}| dt < \infty$$

Existence of Laplace Transform - Region of Convergence *Examples.*

① impulse $\delta(t)$:-

$$\mathcal{L}\{\delta(t)\} = \int_{-\infty}^{\infty} \delta(t) e^{-st} dt = e^{-s(0)} = 1.$$

Roc $\rightarrow$ entire s -plane.



② unit step $u(t)$:-

$$\mathcal{L}\{u(t)\} = \int_{-\infty}^{\infty} u(t) e^{-st} dt$$

$$= \int_0^{\infty} e^{-st} dt = \frac{1}{s} \quad \text{when } \underline{\text{Re}\{s\} > 0}.$$

$$\int_0^{\infty} \underbrace{e^{-\sigma t}} e^{j\omega t} dt$$

1st & 4th Qs

③ $e^{-at} u(t)$:-

$$\mathcal{L}\{e^{-at} u(t)\} = \int_{-\infty}^{\infty} e^{-at} u(t) e^{-st} dt$$

$$= \frac{1}{s+a} \quad \text{when } \text{Re}\{s\} > -a \text{ or } \sigma > -a.$$



Examples

$$\mathcal{L}\{-u(-t)\} = 1/s \quad \text{with ROC: } \sigma < 0.$$

So two different signals can have same $X(s)$ but the ROC is different.